EXERCISE 10.2

Find the derivatives of the following functions with respect to corresponding independent variables:

(1) $f(x) = x - 3\sin x$

(2) $y = \sin x + \cos x$

(3) $f(x) = x \sin x$

(4) $y = \cos x - 2\tan x$

(5) $g(t) = t^3 \cos t$

(6) $g(t) = 4\sec t + \tan t$

(7) $y = e^x \sin x$

(8) $y = \dfrac{\tan x}{x}$

(9) $y = \dfrac{\sin x}{1+\cos x}$

(10) $y = \dfrac{x}{\sin x + \cos x}$

(11) $y = \dfrac{\tan x - 1}{\sec x}$

(12) $y = \dfrac{\sin x}{x^2}$

(13) $y = \tan\theta(\sin\theta + \cos\theta)$

(14) $y = \text{cosec}\, x \,.\, \cot x$

(15) $y = x \sin x \cos x$

(16) $y = e^{-x}.\log x$

(17) $y = (x^2+5)\log(1+x)e^{-3x}$

(18) $y = \sin x°$

(19) $y = \log_{10} x$

(20) Draw the function $f'(x)$ if $f(x) = 2x^2 - 5x + 3$

10.4.2 Examples on Chain Rule

Example 10.8

Find $F'(x)$ if $F(x) = \sqrt{x^2+1}$.

Solution

$$\text{Take } u = g(x) = x^2+1 \text{ and } f(u) = \sqrt{u}$$

$$\therefore F(x) = (fog)(x) = f(g(x))$$

$$\text{Since } f'(u) = \frac{1}{2}u^{-\frac{1}{2}} = \frac{1}{2\sqrt{u}} \text{ and}$$

$$g'(x) = 2x, \text{ we get}$$

$$F'(x) = f'(g(x))g'(x)$$

$$= \frac{1}{2\sqrt{x^2+1}}.2x = \frac{x}{\sqrt{x^2+1}}\,.$$

Example 10.9

Differentiate : (i) $y = \sin(x^2)$ (ii) $y = \sin^2 x$

Solution

(i) The outer function is the sine function and the inner function is the squaring function.

$$\text{Let } u = x^2$$

$$\text{That is, } y = \sin u.$$

$$\text{Therefore, } \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

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